AQ8.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.
Level 1

1 point
Which of these sets form an orthonormal basis of R^3 R3 with respect to the dot product as the inner product in R^3 R3?

- [ ] $\{(1,0,0),(0,1,0)\}$ {(1, 0, 0), (0, 1, 0)}
- [ ] $\{(2,0,0),(0,2,0),(0,0,2)\}$ {(2, 0, 0), (0, 2, 0), (0, 0, 2)}
- [ ] $\{(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}),(\frac{-3}{\sqrt{11}},\frac{1}{\sqrt{11}},\frac{1}{\sqrt{11}}),(\frac{1}{\sqrt{66}},\frac{7}{\sqrt{66}},\frac{-4}{\sqrt{66}})\}$ {(6
√
1, 6
√
1, 6
√
2), (11
√
−3, 11
√
1, 11
√
1), (66
√
1, 66
√
7, 66
√
−4)}
- [ ] $\{(2,1,-1),(-1,-1,1),(1,2,-2)\}$ {(2, 1, −1), (−1, −1, 1), (1, 2, −2)}
- [ ] $\{(\frac{1}{\sqrt{2}},0,\frac{-1}{\sqrt{2}}),(0,1,0),(\frac{1}{\sqrt{2}},0,\frac{1}{\sqrt{2}})\}$ {(2
√
1, 0, 2
√
−1), (0, 1, 0), (2
√
1, 0, 2
√
1)}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\{(\frac{1}{\sqrt{6}},\frac{1}{\sqrt{6}},\frac{2}{\sqrt{6}}),(\frac{-3}{\sqrt{11}},\frac{1}{\sqrt{11}},\frac{1}{\sqrt{11}}),(\frac{1}{\sqrt{66}},\frac{7}{\sqrt{66}},\frac{-4}{\sqrt{66}})\}$ {(6

√
1, 6
√
1, 6
√
2), (11
√
−3, 11
√
1, 11
√
1), (66
√
1, 66
√
7, 66
√
−4)}

$\{(\frac{1}{\sqrt{2}}, 0, \frac{-1}{\sqrt{2}}), (0, 1, 0), (\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}})\}$ {(2
√
1, 0, 2
√
−1), (0, 1, 0), (2
√
1, 0, 2
√
1)}

1 point

Consider two orthogonal vectors a, b in R^2. If $a + b$ and $a - b$ are orthogonal, then choose the correct option.

- ○ $\| a \| = \| b \| = 1$
- ○ $\| a \| = \| b \|$
- ○ $\| a \| = 2 \| b \|$
- ○ $2 \| a \| = \| b \|$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\| a \| = \| b \|$

1 point

Consider $a = (1, 1), b = (1, -1)$. Let $V = Span\{a, b\}$. Choose the correct options by considering the standard inner product (dot product).

- ☐ Vectors a, b form an orthogonal basis for V.
- ☐ Vectors a, b form an orthonormal basis for V.
- ☐

There exist scalar multiples of a, b which form an orthonormal basis

☐

There do not exist scalar multiples of a, b which form an orthonormal basis

No, the answer is incorrect.
Score: 0
Accepted Answers:

Vectors a, b form an orthogonal basis for V .

There exist scalar multiples of a, b which form an orthonormal basis

Level 2

1 point

Choose the set of correct statements.

☐

The determinant of a matrix formed by 3 orthonormal vectors in R^3 is ± 1.

☐

The determinant of a matrix formed by 3 orthonormal vectors in R^3 is 0.

☐

The determinant of a matrix formed by 2 orthonormal vectors in R^2 is ± 1.

☐

The determinant of a matrix formed by 2 orthonormal vectors in R^2 is 0.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The determinant of a matrix formed by 3 orthonormal vectors in R^3 is ± 1.

The determinant of a matrix formed by 2 orthonormal vectors in R^2 is ± 1.

1 point

Consider a system of linear equations:

$$2x_1 + 2x_2 + 7x_3 = b_1$$

$$2x_1 + x_2 - 10x_3 = b_2$$

$$3x_1 - 2x_2 + 2x_3 = b_3.$$

Let A be the coefficient matrix of the given system of linear equations. Let a matrix B contain the column vectors of A, which are normalized by their respective norms, as its columns (i.e. first column vector of A normalized by its norm is the first column of B). Which of the following statements are true ?

☐

The determinant of BB^T is 1.

☐

BB^T is an identity matrix.

☐

BB^T is a scalar matrix.

☐

BB^T is a diagonal matrix.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The determinant of BB^T is 1.

BB^T is an identity matrix.

BB^T is a scalar matrix.
BB^T is a diagonal matrix.

1 point

Choose the correct option(s).

- [ ] The vectors in an orthonormal set are linearly independent.
- [ ] A set of linearly dependent vectors can be orthonormal.
- [ ] A set of linearly independent vectors is always orthonormal.
- [ ] A set of linearly independent vectors is always orthogonal but not orthonormal.
- [ ] The vectors in an orthogonal set are linearly independent.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The vectors in an orthonormal set are linearly independent.
The vectors in an orthogonal set are linearly independent.

Consider the inner product $\langle u, v\rangle = 3u_1v_1 + 2u_2v_2$ where $u = (u_1, u_2), v = (v_1, v_2)$ are vectors in R^2. Answer questions 7 and 8 based on this information.

1 point

Which of the following is an orthonormal basis with respect to the inner product defined above?

- ○ $\{(2,3), (-4,4)\}$
- ○ $\{\frac{1}{\sqrt{30}}(2,3), \frac{1}{\sqrt{80}}(-4,4)\}$
- ○ $\{\frac{1}{\sqrt{13}}(2,3), \frac{1}{\sqrt{32}}(-4,4)\}$
- ○ $\{(2,3), (-3,2)\}$
- ○ $\{\frac{1}{\sqrt{13}}(2,3), \frac{1}{\sqrt{13}}(-3,2)\}$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$\{\frac{1}{\sqrt{30}}(2,3), \frac{1}{\sqrt{80}}(-4,4)\}$

1(−4, 4)}

1 point

Use the orthonormal basis $\{u, v\}${u, v} obtained in question 7 with respect to the defined inner product. Express the vector $(4, 0)$(4, 0) as a linear combination of the basis vectors uu and vv, as $(4, 0) = c_1 u + c_2 v$(4, 0) = c1u + c2v. Which of the following gives the coefficients of the linear combination?

○ $c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}}$c1 = 30

√

24, c2 = 80

√

−48

○ $c_1 = \frac{24}{\sqrt{13}}, c_2 = \frac{-48}{\sqrt{32}}$c1 = 13

√

24, c2 = 32

√

−48

○ $c_1 = \frac{8}{\sqrt{13}}, c_2 = \frac{-16}{\sqrt{32}}$c1 = 13

√

8, c2 = 32

√

−16

○ $c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{48}{\sqrt{80}}$c1 = 30

√

24, c2 = 80

√

48

No, the answer is incorrect.

Score: 0

Accepted Answers:

$c_1 = \frac{24}{\sqrt{30}}, c_2 = \frac{-48}{\sqrt{80}}$c1 = 30

√

24, c2 = 80

√

−48

Check Answers

Your score is: 0/8